

ZIAGEN
Abacavir

QUALITATIVE AND QUANTITATIVE COMPOSITION

Each *ZIAGEN* tablet contains 300 mg of abacavir as abacavir sulfate. The scored tablets are yellow, biconvex, and capsule shaped. They are engraved with GX 623 on both sides.

ZIAGEN oral solution contains 20 mg/ml of abacavir as abacavir sulfate. *ZIAGEN* oral solution is a clear to slightly opalescent yellowish, aqueous solution which may turn into a brown colour over time with a fruity flavor.

CLINICAL INFORMATION

Indications

ZIAGEN is indicated in antiretroviral combination therapy for the treatment of Human Immunodeficiency Virus (HIV) infection in adults and children.

Dosage and Administration

Pharmaceutical Form: Film-coated scored tablets and Oral solution.

Therapy should be initiated by a physician experienced in the management of HIV infection.

ZIAGEN can be taken with or without food.

To ensure administration of the entire dose, the tablet(s) should ideally be swallowed without crushing.

For patients who are unable to swallow tablets, *ZIAGEN* is available as an oral solution. Alternatively, the tablets may be crushed and added to a small amount of semi-solid food or liquid, all of which should be consumed immediately (*see Pharmacokinetics*).

- **Adults, adolescents and children weighing at least 25 kg**

Tablets:

The recommended dose of *ZIAGEN* tablets is 600 mg daily. This may be administered as either 300 mg (one tablet) twice daily or 600 mg (two tablets) once daily.

Oral Solution:

The recommended dose of *ZIAGEN* oral solution is 300 mg (15 ml) twice daily or 600 mg (30 ml) once daily.

- **Children from three months and weighing less than 25 kg**

Tablets:

Children weighing 14 to < 20 kg: one-half of a scored abacavir tablet twice daily.

Children weighing ≥ 20 kg to < 25 kg: one-half of a scored abacavir tablet taken in the morning and one whole tablet taken in the evening.

Children weighing at least 25 kg: the adult dosage of 300 mg twice daily or 600 mg once daily should be taken.

The oral solution may be administered to children weighing less than 14 kg or those who are unable to swallow tablets.

Oral Solution:

The recommended dose is 8 mg/kg twice daily or 16 mg/kg once daily up to a maximum dose of 300 mg twice daily or 600 mg once daily.

- **Children less than three months**

The data available on the use of *ZIAGEN* in this age group are very limited (*see Pharmacokinetics*).

- **Renal impairment**

No dosage adjustment of *ZIAGEN* is necessary in patients with renal dysfunction (*see Pharmacokinetics*). *ZIAGEN* should be avoided in patients with end-stage renal disease.

- **Hepatic impairment**

Abacavir is metabolised primarily by the liver. The recommended dose of *ZIAGEN* in patients with mild hepatic impairment (Child-Pugh score 5 to 6) is 200 mg (10 ml) twice a day. To enable dose reduction *ZIAGEN* oral solution should be used for the treatment of these patients. Pharmacokinetic and safety data on the use of abacavir in patients with moderate and severe hepatic impairment are not available (*see Pharmacokinetics*).

Therefore the use of *ZIAGEN* is not recommended in patients with moderate or severe hepatic impairment, unless the benefit of use outweighs the risk.

Contraindications

ZIAGEN is contraindicated in patients with known hypersensitivity to abacavir or any ingredient of *ZIAGEN* tablets or oral solution.

Warnings and Precautions

Hypersensitivity (*see Adverse Reactions*)

Abacavir is associated with a risk for hypersensitivity reactions (HSR) characterized by fever and/or rash with other symptoms indicating multi-organ involvement. HSR can be life-threatening, and in rare cases fatal, when not managed appropriately. The risk for abacavir HSR to occur is significantly increased for patients who test positive for the *HLA-B*5701* allele. However, abacavir HSRs have been reported at a lower frequency in patients who do not carry this allele.

The following should be adhered to:

- Testing for *HLA-B*5701* status should be considered before initiating abacavir treatment and also before re-starting abacavir treatment in patients of unknown *HLA-B*5701* status who have previously tolerated abacavir.
- *ZIAGEN* is not recommended for use in patients with the *HLA-B*5701* allele, or in patients who have had a suspected abacavir HSR while taking any other medicinal product containing abacavir (e.g. *KIVEXA*,) regardless of *HLA-B*5701* status.
- Each patient should be reminded to read the Patient Leaflet included in the *ZIAGEN* pack. They should be reminded of the importance of removing the Alert Card included in the pack, and keeping it with them at all times.
- In any patient treated with *ZIAGEN*, the clinical diagnosis of suspected hypersensitivity reaction must remain the basis of clinical decision making.
- ***ZIAGEN* must be stopped without delay, even in the absence of the *HLA-B*5701* allele, if a HSR is suspected. Delay in stopping treatment with *ZIAGEN* after the onset of hypersensitivity may result in a life-threatening reaction.**
- Patients who have experienced a hypersensitivity reaction should be instructed to dispose of their remaining *ZIAGEN* tablets in order to avoid restarting abacavir.
- **Restarting abacavir containing products following a suspected abacavir HSR can result in a prompt return of symptoms within hours, and may include life-threatening hypotension and death.**
- **Regardless of a patient's *HLA-B*5701* status, if therapy with any abacavir containing product has been discontinued for any reason and restarting abacavir**

therapy is under consideration, the reason for discontinuation must be established. If HSR cannot be ruled out, *ZIAGEN* or any other medicinal product containing abacavir (e.g. *KIVEXA*) must not be restarted.

- If a hypersensitivity reaction is ruled out, patients may restart *ZIAGEN*. Rarely, patients who have stopped abacavir for reasons other than symptoms of HSR have also experienced life-threatening reactions within hours of re- initiating abacavir therapy (*see Section 4.8 Description of selected adverse reactions*). Patients must be made aware that HSR can occur with reintroduction of *ZIAGEN* or any other medicinal product containing abacavir (e.g. *KIVEXA*) and that reintroduction of *ZIAGEN* or any other medicinal product containing abacavir (e.g. *KIVEXA*) should be undertaken only if medical care can be readily accessed.

Clinical Description of abacavir HSR:

Abacavir HSR has been well characterised through clinical studies and during post marketing follow-up. Symptoms usually appeared within the first six weeks (median time to onset 11 days) of initiation of treatment with abacavir, **although these reactions may occur at any time during therapy.**

Almost all HSR to abacavir include fever and/or rash as part of the syndrome.

Other signs and symptoms that have been observed as part of abacavir HSR include respiratory and gastrointestinal symptoms, **which may lead to misdiagnosis of HSR as respiratory disease (pneumonia, bronchitis, pharyngitis), or gastroenteritis** (*see Adverse Reactions, Description of Selected Adverse Reactions*). The symptoms related to HSR worsen with continued therapy and can be life threatening. These symptoms usually resolve upon discontinuation of *ZIAGEN*.

- **Lactic Acidosis/Severe Hepatomegaly with Steatosis**

Lactic acidosis and severe hepatomegaly with steatosis, including fatal cases, have been reported with the use of antiretroviral nucleoside analogues either alone or in combination, including abacavir. A majority of these cases have been in women.

Clinical features which may be indicative of the development of lactic acidosis include generalised weakness, anorexia, and sudden unexplained weight loss, gastrointestinal symptoms and respiratory symptoms (dyspnoea and tachypnoea).

Caution should be exercised when administering *ZIAGEN*, particularly to those with known risk factors for liver disease. Treatment with *ZIAGEN* should be suspended in any patient who develops clinical or laboratory findings suggestive of lactic acidosis with or without hepatitis (which may include hepatomegaly and steatosis even in the absence of marked transaminase elevations).

- ***Serum lipids and blood glucose***

Serum lipid and blood glucose levels may increase during antiretroviral therapy. Disease control and lifestyle changes may also be contributing factors. Consideration should be given to the measurement of serum lipids and blood glucose. Lipid disorders should be managed as clinically appropriate.

- **Immune Reconstitution Syndrome**

In HIV-infected patients with severe immune deficiency at the time of initiation of anti-retroviral therapy (ART), an inflammatory reaction to asymptomatic or residual opportunistic infections may arise and cause serious clinical conditions, or aggravation of symptoms. Typically, such reactions have been observed within the first few weeks or months of initiation of ART. Relevant examples are cytomegalovirus retinitis, generalised and/or focal mycobacterial infections and *Pneumocystis jiroveci* pneumonia (often referred to as PCP). Any inflammatory symptoms must be evaluated without delay and treatment initiated when necessary. Autoimmune disorders (such as Graves' disease, polymyositis and Guillain-Barre syndrome) have also been reported to occur in the setting of immune reconstitution, however, the time to onset is more variable, and can occur many months after initiation of treatment and sometimes can be an atypical presentation.

- **Opportunistic infections**

Patients receiving *ZIAGEN* or any other antiretroviral therapy may still develop opportunistic infections and other complications of HIV infection. Therefore patients should remain under close clinical observation by physicians experienced in the treatment of these associated HIV diseases.

- **Myocardial Infarction**

Several observational, epidemiological studies have reported an association with abacavir use and the risk of myocardial infarction. Meta-analyses of randomised controlled trials have observed no excess risk of myocardial infarction with abacavir use. To date, there is no established biological mechanism to explain a potential increase in risk. In totality the available data from observational studies and from controlled clinical trials show inconsistency and therefore the evidence for a causal relationship between abacavir treatment and the risk of myocardial infarction is inconclusive.

As a precaution the underlying risk of coronary heart disease should be considered when prescribing antiretroviral therapies, including abacavir, and action taken to minimize all modifiable risk factors (e.g. hypertension, hyperlipidaemia, diabetes mellitus and smoking).

Oral solution:

ZIAGEN oral solution contains sorbitol which may cause abdominal pain and diarrhoea. Sorbitol is metabolised to fructose and is therefore unsuitable for patients who have hereditary fructose intolerance.

Interactions

The potential for drug interactions involving abacavir is low. *In vitro* studies have shown that abacavir has potential to inhibit cytochrome P450 1A1 (CYP1A1). Abacavir shows limited potential to inhibit metabolism mediated by the CYP3A4 enzyme. It has also been shown *in vitro* not to interact with drugs that are metabolised by CYP2C9 or CYP2D6 enzymes. Induction of hepatic metabolism has not been observed in clinical studies. Therefore, there is little potential for drug interactions with antiretroviral protease inhibitors and other drugs metabolised by major CYP enzymes. Clinical studies have shown that there are no clinically significant interactions between abacavir, zidovudine and lamivudine.

Effect of Abacavir on the Pharmacokinetics of Other Agents

In vitro, abacavir demonstrates no or weak inhibition of the drug transporters organic anion transporter 1B1 (OATP1B1), OATP1B3, breast cancer resistance protein (BCRP) or P-glycoprotein (Pgp) and minimal inhibition of organic cation transporter 1 (OCT1), OCT2 and multidrug and toxin extrusion protein 2-K (MATE2-K). Abacavir is therefore not expected to affect the plasma concentrations of drugs that are substrates of these drug transporters.

Abacavir is an inhibitor of MATE1 *in vitro*, however abacavir has low potential to affect the plasma concentrations of MATE1 substrates at therapeutic drug exposures (up to 600 mg).

Effect of Other Agents on the Pharmacokinetics of Abacavir

In vitro, abacavir is not a substrate of OATP1B1, OATP1B3, OCT1, OCT2, OAT1, MATE1, MATE2-K, Multidrug resistance-associated protein 2 (MRP2) or MRP4, therefore drugs that modulate these transporters are not expected to affect abacavir plasma concentrations.

Although abacavir is a substrate of BCRP and Pgp *in vitro*, clinical studies demonstrate no clinically significant changes in abacavir pharmacokinetics when co-administered with lopinavir/ritonavir (Pgp and BCRP inhibitors).

Interactions relevant to abacavir

Ethanol: The metabolism of abacavir is altered by concomitant ethanol resulting in an increase in AUC of abacavir of about 41%. Given the safety profile of abacavir these findings are not considered clinically significant. Abacavir has no effect on the metabolism of ethanol.

Methadone: In a pharmacokinetic study, coadministration of 600 mg *ZIAGEN* twice daily with methadone showed a 35% reduction in abacavir C_{max} and a 1 h delay in t_{max} , but the AUC was unchanged. The changes in abacavir pharmacokinetics are not considered clinically relevant. In this study abacavir increased the mean methadone systemic clearance by 22 %. This change is not considered clinically relevant for the majority of patients, however, occasionally methadone re-titration may be required.

Retinoids: Retinoid compounds such as isotretinoin, are eliminated via alcohol dehydrogenase. Interaction with abacavir is possible but has not been studied.

Riociguat: *In vitro*, abacavir inhibits CYP1A1. Concomitant administration of a single dose of riociguat (0.5 mg) to HIV patients receiving the combination of abacavir/dolutegravir/lamivudine (600mg/50mg/300mg once daily) led to an approximately three-fold higher riociguat AUC_(0-∞) when compared to historical riociguat AUC_(0-∞) reported in healthy subjects. Riociguat dose may need to be reduced, consult the riociguat product labeling for dosing recommendations.

Pregnancy and Lactation

Pregnancy

Abacavir has been evaluated in the Antiretroviral Pregnancy Registry in over 2000 women during pregnancy and postpartum. Available human data from the Antiretroviral Pregnancy Registry do not show an increased risk of major birth defects for abacavir compared to the background rate (*see Clinical Studies*). However, there are no adequate and well-controlled trials in pregnant women and the safe use of *ZIAGEN* in human pregnancy has not been established. Abacavir has been associated with findings in animal reproductive studies (*see Non-Clinical Information*). Therefore administration of *ZIAGEN* in pregnancy should be considered only if the benefit to the mother outweighs the possible risk to the foetus.

There have been reports of mild, transient elevations in serum lactate levels, which may be due to mitochondrial dysfunction, in neonates and infants exposed *in utero* or peri-partum to NRTIs. The clinical relevance of transient elevations in serum lactate is unknown. There have also been very rare reports of developmental delay, seizures and other neurological disease. However, a causal relationship between these events and NRTI exposure *in utero* or peri-partum has not been established. These findings do not affect current recommendations to use antiretroviral therapy in pregnant women to prevent vertical transmission of HIV.

Lactation

Health experts recommend that where possible HIV infected women do not breast feed their infants in order to avoid transmission of HIV. In settings where formula feeding is not feasible, local official lactation and treatment guidelines should be followed when considering breast feeding during antiretroviral therapy.

In a study after repeat oral administration of 300 mg abacavir twice daily (given as *TRIZIVIR*), the breast milk: maternal plasma ratio was 0.9. Most infants (8 out of 9) had non-detectable levels of abacavir (assay sensitivity 16ng/mL). Intracellular carbovir triphosphate (active metabolite of abacavir) levels in breastfed infants were not measured therefore the clinical relevance of the serum concentrations of the parent compound measured is unknown.

Effects on Ability to Drive and Use Machines

No currently available data suggest that *ZIAGEN* affects the ability to drive or operate machinery.

Adverse Reactions

For many of the other adverse events reported, it is unclear whether they are related to *ZIAGEN*, to the wide range of medicinal products used in the management of HIV disease or as a result of the disease process.

Many of those listed below (nausea, vomiting, diarrhoea, fever, fatigue, rash) occur commonly as part of *ZIAGEN* hypersensitivity. Therefore, patients with any of these symptoms should be carefully evaluated for the presence of this hypersensitivity reaction. If *ZIAGEN* has been discontinued in patients due to experiencing any one of these symptoms and a decision is made to restart *ZIAGEN*, this should be done only under direct medical supervision (*see Warnings and Precautions - "Special considerations following an interruption of ZIAGEN therapy"*).

The majority of the adverse reactions listed below have not been treatment limiting. The following convention has been used for their classification: very common (more than 1/10), common (more than 1/100, less than 1/10), uncommon (more than 1/1,000, less than 1/100), rare (more than 1/10,000, less than 1/1,000) very rare (less than 1/10,000).

Clinical Trial Data

Metabolism and nutrition disorders

Common: anorexia.

Nervous system disorders

Common: headache.

Gastrointestinal disorders

Common: nausea, vomiting, diarrhoea.

General disorders and administration site disorders

Common: fever, lethargy, fatigue.

In controlled clinical studies laboratory abnormalities related to *ZIAGEN* treatment were uncommon, with no differences in incidence observed between *ZIAGEN* treated patients and the control arms.

Paediatric population

The safety database to support abacavir once daily dosing in paediatric patients comes from the

ARROW Trial (COL105677) in which 669 HIV-1 infected paediatric subjects received abacavir and lamivudine either once or twice daily (*see Clinical Studies*). No additional safety issues have been identified in paediatric subjects receiving either once or twice daily dosing compared to adults.

Postmarketing Data

Metabolism and nutrition disorders

Common: hyperlactataemia.

Rare: lactic acidosis (*see Warnings and Precautions*).

Gastrointestinal disorders

Rare: pancreatitis has been reported, but a causal relationship to *ZIAGEN* treatment is uncertain.

Skin and subcutaneous tissue disorders

Common: rash (without systemic symptoms).

Very rare: erythema multiforme, Stevens-Johnson syndrome and toxic epidermal necrolysis.

Description of Selected Adverse Reactions

Hypersensitivity (see also Warnings and Precautions):

Abacavir hypersensitivity reaction (HSR) has been identified as a common adverse reaction with abacavir therapy. The signs and symptoms of this hypersensitivity reaction are listed below. These have been identified either from clinical studies or post marketing surveillance. Those reported in **at least 10% of patients** with a hypersensitivity reaction are in bold text.

Almost all patients developing hypersensitivity reactions will have fever and/or rash (usually maculopapular or urticarial) as part of the syndrome, however, reactions have occurred without rash or fever. Other key symptoms include gastrointestinal, respiratory or constitutional symptoms such as lethargy and malaise.

Skin: **Rash** (usually maculopapular or urticarial)

Gastrointestinal tract: **Nausea, vomiting, diarrhoea, abdominal pain, mouth ulceration**

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Respiratory tract:	Dyspnoea, cough , sore throat, adult respiratory distress syndrome, respiratory failure
Miscellaneous:	Fever, fatigue, malaise , oedema, lymphadenopathy, hypotension, conjunctivitis, anaphylaxis
Neurological/Psychiatry:	Headache , paraesthesia
Haematological:	Lymphopenia
Liver/pancreas:	Elevated liver function tests , hepatic failure
Musculoskeletal:	Myalgia , rarely myolysis, arthralgia, elevated creatine phosphokinase
Urology:	Elevated creatinine, renal failure

Restarting abacavir following an abacavir HSR results in a prompt return of symptoms within hours. This recurrence of the HSR is usually more severe than on initial presentation, and may include life-threatening hypotension and death. Reactions have also occurred infrequently after restarting abacavir in patients who had only one of the key symptoms of hypersensitivity (see above) prior to stopping abacavir; and on very rare occasions have also been seen in patients who have restarted therapy with no preceding symptoms of a HSR (i.e., patients previously considered to be abacavir tolerant).

For details of clinical management in the event of a suspected abacavir HSR see Warnings and Precautions.

Overdose

Single doses up to 1200 mg and daily doses up to 1800 mg of abacavir have been administered to patients in clinical studies. No unexpected adverse reactions were reported. The effects of higher doses are not known.

If overdosage occurs the patient should be monitored for evidence of toxicity (*see Adverse Reactions*), and standard supportive treatment applied as necessary. It is not known whether abacavir can be removed by peritoneal dialysis or haemodialysis.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamics

Pharmacotherapeutic group - nucleoside analogue, ATC Code: J05A F06.

Mechanism of action

Abacavir is a nucleoside analogue reverse transcriptase inhibitor. It is a potent selective inhibitor of HIV-1 and HIV-2, including HIV-1 isolates with reduced susceptibility to zidovudine, lamivudine, zalcitabine, didanosine or nevirapine. Abacavir is metabolised intracellularly to the active moiety, carbovir 5'-triphosphate (TP). *In vitro* studies have demonstrated that its mechanism of action in relation to HIV is inhibition of the HIV reverse transcriptase enzyme, an event which results in chain termination and interruption of the viral replication cycle.

The antiviral activity of abacavir in cell culture was not antagonized when combined with the nucleoside reverse transcriptase inhibitors (NRTIs) didanosine, emtricitabine, lamivudine, stavudine, tenofovir, zalcitabine or zidovudine, the non-nucleoside reverse transcriptase inhibitor (NNRTI) nevirapine, or the protease inhibitor (PI) amprenavir.

Pharmacodynamic effects

In a study of 20 HIV-infected patients receiving *ZIAGEN* 300 mg twice daily, with only one 300 mg dose taken prior to the 24 hours sampling period, the geometric mean terminal carbovir-TP intracellular half-life at steady-state was 20.6 hours, compared to the geometric mean abacavir plasma half-life in this study of 2.6 hours. The steady state pharmacokinetic properties of *ZIAGEN* 600 mg once daily was compared to *ZIAGEN* 300 mg twice daily in a crossover study in 27 HIV-infected patients. Intracellular carbovir triphosphate exposures in peripheral blood mononuclear cells were higher for *ZIAGEN* 600 mg once daily with respect to AUC_{24,ss} (32 %, higher), C_{max} 24,ss (99% higher) and trough values (18% higher), compared to the 300 mg twice daily regimen. These data support the use of *ZIAGEN* 600 mg once daily for the treatment of HIV infected patients. Additionally, the efficacy and safety of *ZIAGEN* given once daily has been demonstrated in a pivotal clinical study (CNA30021- *see Clinical Studies*).

Abacavir-resistant isolates of HIV-1 have been selected *in vitro* and are associated with specific genotypic changes in the reverse transcriptase (RT) codon region (codons M184V, K65R, L74V and Y115F). Viral resistance to abacavir develops relatively slowly *in vitro* and *in vivo*, requiring multiple mutations to reach an eight-fold increase in IC₅₀ over wild-type virus, which may be a clinically relevant level.

Isolates resistant to abacavir may also show reduced sensitivity to lamivudine, zalcitabine and/or didanosine, but remain sensitive to zidovudine and stavudine. Cross resistance between abacavir and protease inhibitors or non nucleoside reverse transcriptase inhibitors is unlikely. Treatment failure following initial therapy with abacavir, lamivudine and zidovudine is mainly associated with the M184V alone, thus maintaining many therapeutic options for a second line regimen.

Abacavir penetrates the cerebrospinal fluid (CSF) (*see Pharmacokinetics*), and has been shown to reduce HIV-1 RNA levels in the CSF. In combination with other antiretrovirals it may have a role in the prevention of HIV related neurological complications, and may delay the development of resistance in this sanctuary site.

Pharmacokinetics

Absorption

Abacavir is rapidly and well absorbed following oral administration. The absolute bioavailability of oral abacavir in adults is about 83%. Following oral administration, the mean time ($t_{\max}$) to maximal serum concentrations of abacavir is about 1.5 hours for the tablet formulation and about 1 hour for the solution formulation.

There are no differences observed between the AUC for the tablet or solution. At a dosage of 300 mg twice daily, the mean steady state $C_{\max}$ of abacavir from tablet administration was 3.00 micrograms/ml, and the mean AUC over a dosing interval of 12 h was 6.02 micrograms.h/ml (daily AUC of approximately 12.0 micrograms.h/ml). The $C_{\max}$ value for the oral solution is slightly higher than the tablet. After a 600 mg *ZIAGEN* tablet dose, the mean abacavir $C_{\max}$ was approximately 4.26 micrograms/ml and the mean AUC_{∞} was 11.95 micrograms.h/ml.

Food delayed absorption and decreased $C_{\max}$ but did not affect overall plasma concentrations (AUC). Therefore *ZIAGEN* can be taken with or without food.

Administration of crushed tablets with a small amount of semi-solid food or liquid would not be expected to have an impact on the pharmaceutical quality, and would therefore not be expected to alter the clinical effect. This conclusion is based on the physiochemical and pharmacokinetic characteristics of the active ingredient and the *in vitro* dissolution behaviour of abacavir tablets in water, assuming that the patient crushes and transfers 100% of the tablet and ingests immediately.

Distribution

Following i.v. administration, the apparent volume of distribution was about 0.8 l/kg, indicating that abacavir penetrates freely into body tissues.

Studies in HIV infected patients have shown good penetration of abacavir into the cerebrospinal fluid (CSF), with a CSF to plasma AUC ratio of between 30 to 44%. In a Phase I pharmacokinetic study, the penetration of abacavir into the CSF was investigated following administration of *ZIAGEN* 300 mg twice a day. The mean concentration of abacavir achieved in the CSF 1.5 hours post dose was 0.14 micrograms/ml. In a further pharmacokinetic study of 600 mg twice a day, the CSF concentration of abacavir increased over time, from approximately 0.13 micrograms/ml at 0.5 to 1 hour after dosing, to approximately 0.74 micrograms/ml after 3 to 4 hours. While peak concentrations may not have been attained by 4 hours, the observed values are 9-fold greater than the IC₅₀ of abacavir of 0.08 micrograms/ml or 0.26 micromoles.

Plasma protein binding studies *in vitro* indicate that abacavir binds only low to moderately (circa

49%) to human plasma proteins at therapeutic concentrations. This indicates a low likelihood for drug interactions through plasma protein binding displacement.

Metabolism

Abacavir is primarily metabolised by the liver with less than 2% of the administered dose being renally excreted, as unchanged compound. The primary pathways of metabolism in man are by alcohol dehydrogenase and by glucuronidation to produce the 5'-carboxylic acid and 5'-glucuronide which account for about 66% of the administered dose. These metabolites are excreted in the urine.

Elimination

The mean half-life of abacavir is about 1.5 hours. Following multiple oral doses of *ZIAGEN* 300 mg twice a day there is no significant drug accumulation. Elimination of abacavir is via hepatic metabolism with subsequent excretion of metabolites primarily in the urine. The metabolites and unchanged abacavir account for about 83% of the administered *ZIAGEN* dose in the urine the remainder is eliminated in the faeces.

Special Patient Populations

- **Children**

Abacavir is rapidly and well absorbed from oral solution and tablet formulations administered to children. Plasma abacavir exposure has been shown to be the same for both formulations when administered at the same dose. Children receiving abacavir oral solution according to the recommended dosage regimen achieve plasma abacavir exposure similar to adults. Children receiving abacavir oral tablets according to the recommended dosage regimen achieve higher plasma abacavir exposure than children receiving oral solution because higher mg/kg doses are administered with the tablet formulation (*see Dosage and Administration*). Paediatric pharmacokinetic studies have demonstrated that once daily dosing provides equivalent AUC₀₋₂₄ to twice daily dosing of the same total daily dose for both oral solution and tablet formulations.

There are insufficient safety data to recommend the use of *ZIAGEN* in infants less than 3 months old. The limited data available indicate that a dose of 2 mg/kg in neonates less than 30 days old provides similar or greater AUCs, compared to the 8 mg/kg dose administered to older children.

- **Elderly**

The pharmacokinetics of abacavir have not been studied in patients over 65 years of age. When treating elderly patients consideration needs to be given to the greater frequency of decreased hepatic, renal and cardiac function, and concomitant disease or other drug therapy.

- **Renally impaired**

Abacavir is primarily metabolised by the liver with approximately 2% of abacavir excreted

unchanged in the urine. The pharmacokinetics of abacavir in patients with end-stage renal disease is similar to patients with normal renal function. Therefore no dosage reduction is required in patients with renal impairment.

- **Hepatically impaired**

Abacavir is metabolised primarily by the liver. The pharmacokinetics of abacavir have been studied in patients with mild hepatic impairment (Child-Pugh score 5 to 6). The results showed that there was a mean increase of 1.89 fold in the abacavir AUC, and 1.58 fold in the half-life of abacavir. The AUCs of the metabolites were not modified by the liver disease. However, the rates of formation and elimination of these were decreased.

In order to achieve exposures that are within the therapeutic range of patients without liver disease, patients with mild hepatic impairment should receive 200 mg *ZIAGEN* twice daily. The pharmacokinetics have not been studied in patients with moderate or severe hepatic impairment, therefore *ZIAGEN* is not recommended in these patient groups.

Clinical Studies

In a double-blind clinical study over 48 weeks in treatment naïve adult patients, the combination of abacavir, lamivudine and zidovudine showed an equivalent antiviral effect to the combination with indinavir, lamivudine and zidovudine in the primary analysis of efficacy. In a secondary analysis of patients with baseline plasma HIV-1 RNA levels above 100,000 copies per ml, patients receiving the combination containing indinavir had a superior response. Patients with baseline plasma HIV-1 RNA below 100,000 copies per ml had an equivalent response to both treatments.

A once daily regimen of abacavir and lamivudine was investigated in a multicentre, double-blind, controlled study (CNA30021) of 770 HIV-infected, therapy-naïve adults. They were randomised to receive either *ZIAGEN* 600 mg once daily or 300 mg twice daily, both in combination with lamivudine 300 mg once daily and efavirenz 600 mg once daily. Patients were stratified at baseline based on plasma HIV-1 RNA less than or equal to 100,000 copies/ml or greater than 100,000 copies/ml. The duration of double-blind treatment was at least 48 weeks. The results are summarised in the table below.

Virological Response Based on Plasma HIV-1 RNA less than 50 copies/ml at Week 48 ITT-Exposed Population

Populations	ABC once/day + 3TC + EFV (N = 384)	ABC twice/day + 3TC + EFV (N = 386)
Sub-group by baseline RNA		
Less than or equal to 100,000 copies/ml	141/217 (65%)	145/217 (67%)
Greater than 100,000 copies/ml	112/167 (67%)	116/169 (69%)
Total population	253/384 (66%)	261/386 (68%)

The abacavir once daily group was demonstrated to be non-inferior when compared to the twice daily group in the overall and base-line viral load sub-groups. The incidence of adverse events reported was similar in the two treatment groups.

Genotypic analysis was attempted for all subjects with virologic failure (confirmed HIV RNA greater than 50 copies/ml). There was a low overall incidence of virologic failure in both the once and twice daily treatment groups (10% and 8%, respectively). Additionally genotyping was restricted to samples with plasma HIV-1 RNA greater than 500 copies/ml. These factors resulted in a small sample size. Therefore no firm conclusions could be drawn regarding differences in treatment emergent mutations between the two treatment groups. Reverse transcriptase amino acid residue 184 was consistently the most frequent position for NRTI resistance-associated mutations (M184V or M184I). The second most frequent mutation was L74V. Mutations Y115F and K65R were uncommon.

A randomised comparison of a regimen including once daily vs twice daily dosing of abacavir and lamivudine was undertaken within a randomised, multicentre, controlled study of HIV-infected, paediatric patients. 1206 paediatric patients aged 3 months to 17 years enrolled in the ARROW Trial (COL105677) and were dosed according to the weight - band dosing recommendations in the World Health Organisation treatment guidelines (Antiretroviral therapy of HIV infection in infants and children, 2006). After 36 weeks on a regimen including twice daily abacavir and lamivudine, 669 eligible subjects were randomised to either continue twice daily dosing or switch to once daily abacavir and lamivudine for at least 96 weeks. The results are summarised in the table below:

Virological Response Based on Plasma HIV-1 RNA less than 80 copies/ml at Week 48 and Week 96 in the Once Daily versus Twice Daily abacavir + lamivudine randomisation of ARROW (Observed Analysis)

	Twice Daily n/N (%)	Once Daily n/N (%)
Week 0 (After ≥36 Weeks on Treatment)		
Plasma HIV-1 RNA <80 c/mL	250/331 (76)	237/335 (71)
Risk difference (once daily-twice daily)	-4.8% (95% CI -11.5% to +1.9%), p=0.16	
Week 48		
Plasma HIV-1 RNA <80 c/mL	242/331 (73)	236/330 (72)
Risk difference (once daily-twice daily)	-1.6% (95% CI -8.4% to +5.2%), p=0.65	
Week 96		
Plasma HIV-1 RNA <80 c/mL	234/326 (72)	230/331 (69)
Risk difference (once daily-twice daily)	-2.3% (95% CI -9.3% to +4.7%), p=0.52	

The abacavir/lamivudine once daily dosing group was demonstrated to be non-inferior to the twice daily group according to the pre-specified non-inferiority margin of -12%, for the primary endpoint of <80 c/mL at Week 48 as well as at Week 96 (secondary endpoint) and all other thresholds tested (<200c/mL, <400c/mL, <1000c/mL), which all fell well within this non-inferiority margin. Subgroup analyses testing for heterogeneity of once vs twice daily demonstrated no significant effect of sex, age, or viral load at randomisation. Conclusions supported non-inferiority regardless of analysis method.

In a study comparing unblinded NRTI combinations (with or without blinded nelfinavir) in children, a significantly greater proportion treated with abacavir and lamivudine (73%) or abacavir and zidovudine (70%) had HIV-1 RNA less than or equal to 400 copies/ml at 24 weeks, compared with those treated with lamivudine and zidovudine (44%). In children with extensive antiretroviral exposure, a modest but sustained effect of the combination of abacavir, lamivudine and zidovudine was observed.

In therapy experienced patients, the degree of benefit from the addition of abacavir will depend on the nature and duration of prior therapy which may have selected for cross resistance to abacavir.

Antiretroviral Pregnancy Registry

The Antiretroviral Pregnancy Registry has received prospective reports of over 2,000 exposures to abacavir during pregnancy resulting in live birth. These consist of over 800 exposures during the first trimester, over 1,100 exposures during the second/third trimester and included 27 and 32 birth defects respectively. The prevalence (95% CI) of defects in the first trimester was 3.1% (2.0, 4.4%) and in the second/third trimester, 2.7% (1.9, 3.9%). Among pregnant women in the reference population, the background rate of birth defects is 2.7%. There was no association between abacavir and overall birth defects observed in the Antiretroviral Pregnancy Registry.

Non-Clinical Information

- **Carcinogenesis, mutagenesis**

Abacavir was not mutagenic in bacterial tests but showed activity *in vitro* in the human lymphocyte chromosome aberration assay, the mouse lymphoma assay, and the *in vivo* micronucleus test. This is consistent with the known activity of other nucleoside analogues. These results indicate that abacavir is a weak clastogen both *in vitro* and *in vivo* at high test concentrations.

Carcinogenicity studies with orally administered abacavir in mice and rats showed an increase in the incidence of malignant and non-malignant tumours. Malignant tumours occurred in the preputial gland of males and the clitoral gland of females of both species, and in the liver, urinary bladder, lymph nodes and the subcutis of female rats.

The majority of these tumours occurred at the highest abacavir dose of 330 mg/kg/day in mice and 600 mg/kg/day in rats. These dose levels were equivalent to 24 to 32 times the expected systemic exposure in humans. The exception was the preputial gland tumour which occurred at a dose of 110 mg/kg. This is equivalent to six times the expected human systemic exposure. There is no structural counterpart for this gland in humans. While the carcinogenic potential in humans is unknown, these data suggest that a carcinogenic risk to humans is outweighed by the potential clinical benefit.

- **Reproductive toxicology**

Placental transfer of abacavir and/or its related metabolites has been shown to occur in animals. Evidence of toxicity to the developing embryo and fetuses occurred only in rats at maternally toxic doses of 500 mg/kg/day and above. This dose is equivalent to 32 to 35 times human therapeutic exposure based on AUC. The findings included foetal oedema, variations and malformations, resorptions, decreased foetal body weight and an increase in still births. The dose at which there were no effects on pre or post natal development was 160 mg/kg/day. This dose is equivalent to an exposure of about 10 times that in humans. Similar findings were not observed in

rabbits.

A fertility study in the rat has shown that doses up to 500 mg/kg of abacavir had no effect on male or female fertility.

- **Animal toxicology and/or pharmacology**

Mild myocardial degeneration in the heart of mice and rats was observed following administration of abacavir for two years. The systemic exposures were equivalent to 7 to 24 times the expected systemic exposure in humans. The clinical relevance of this finding has not been determined.

PHARMACEUTICAL INFORMATION

List of Excipients

Tablet Core

Microcrystalline cellulose
Sodium starch glycollate (Type A)
Magnesium stearate
Colloidal anhydrous silica

Tablet Coating

Opadry yellow YS-1-12789-A
Purified water

Oral Solution

Sorbitol (*See Warning and Precautions*)
Saccharin sodium
Sodium citrate
Citric acid anhydrous
Methyl parahydroxybenzoate (E218)
Propyl parahydroxybenzoate (E216)
Propylene glycol
Artificial strawberry and banana flavour
Hydrochloric acid, dilute and/or solution of sodium hydroxide
Purified water

Shelf Life

The expiry date is indicated on the packaging.

Storage

Tablets

Do not store above 30°C.

Oral solution

Do not store above 30°C.

Discard oral solution 2 months after first opening.

Nature and Contents of Container

ZIAGEN tablets are available in polyvinyl chloride/foil or polyvinyl chloride/child resistant foil blister packs containing 60 tablets.

ZIAGEN oral solution is supplied in high density polyethylene bottles with child-resistant closures, containing 240 ml of oral solution. A 10 ml polypropylene oral dosing syringe and a polyethylene adapter are also included in the pack.

Incompatibilities

Not known.

Use and Handling

An oral dosing syringe is provided for accurate measurement of the prescribed dose of *ZIAGEN* oral solution. Instructions for use are included in the pack.

Discard *ZIAGEN* oral solution two months after first opening.

Not all presentations are available in every country.

Manufacturer

Ziagen Tablets

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ul. Grunwaldzka 189
60-322 Poznań
Poland

Ziagen Oral Solution

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Version number: 06

Reference: GDS33/IPI14

Date of local revision: 15 March 2023